



Andrei-Claudiu Roibu

Applied AI PhD • Software Engineer

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Profile

Machine learning engineer (Oxford PhD) working in a customer-facing role at Synthara, delivering PoCs and supporting benchmarking on embedded hardware for AI and signal processing workloads. Previously worked in pharma and aerospace, building applied deep learning and data science pipelines. I focus on delivering practical, end-to-end solutions that bridge technical expertise with project delivery and clear stakeholder communication.

Education

PhD in Applied AI and Biomedical Data Science

Oct 2019 - Aug 2023

University of Oxford

Oxford, United Kingdom

- **Thesis:** Deep Learning Approaches to Multimodal MRI Brain Age Estimation ([link](#) / PhD awarded Nov 2024)

MEng in Aerospace Engineering (First-Class Honours, Valedictorian)

Sep 2014 - Jun 2019

University of Sheffield

Sheffield, United Kingdom

- **Thesis:** Design of Artificial Intelligence Agents for Games using Deep Reinforcement Learning ([link](#))

Experience

Software Engineer (Customer Solutions / Field Applications - [link](#))

Sep 2023 - Present

Synthara AG

Zürich, Switzerland

Synthara designs high-performance, energy-efficient in-memory computing (IMC) chips for embedded devices.

- Engineered an automated benchmarking pipeline (GCP, MongoDB, CI), reducing turnaround from weeks to hours.
- Benchmarked 50+ workloads (ML, LLM, DSP), mapping new layers to IMC matrix-algebra primitives.
- Productised a transaction-accurate emulator into a digital twin, owning requirements, validation and hardening.
- Established a spec-driven agent workflow for code implementation and independent review with human sign-off.
- Supported 15+ pre-sales activities, translating customer requirements into KPI-driven design-space explorations.
- Shaped the applications roadmap, drove Lean Agile adoption, and managed and mentored junior engineers.
- Supported business development via technical workshops and collateral, helping generate B2B and VC leads.

Doctoral Researcher ([link](#))

Oct 2019 - Aug 2023

University of Oxford and F. Hoffmann-La Roche AG

Oxford, United Kingdom & Basel, Switzerland

Developed 3D multimodal deep learning and data analysis pipelines, linking complex brain ageing processes to biological, environmental, and lifestyle factors and supporting research into neurodegenerative diseases.

- Built a deep learning HPC pipeline, using multimodal 3D MRIs from 30k UK Biobank subjects to predict brain age.
- Developed 3D CNNs and SWIN Vision Transformers, optimising for accuracy, robustness, and explainability.
- Corrected for variance, bias, and confounds, finding 100+ factors associated with healthy ageing deviations.
- Demonstrated that ElasticNet ensembling surpasses deep fusion for multimodal predictions, indicating trade-offs.
- Leveraged PCA-ICA methods on predictions to uncover independent modes of brain ageing within a population.
- Engineered deep autoencoders to predict brain function from structural maps, outperforming PCA/PLS baselines.
- Collaborated with Roche pRED to translate research into prototype models supporting early drug discovery.

Product Manager / Software Engineer (Part-Time) ([link](#))

Oct 2019 - Oct 2020

GE HealthCare UK Plc.

Oxford, United Kingdom

Our team built a local tool providing editable brain MRI segmentations of problematic neurodegenerative regions.

- Coordinated a small agile team building [Paint4Brains](#), translating clinician requirements into a prioritised backlog.

- Integrated [*QuickNat*](#), improving Dice by 11% and CPU-laptop inference speed by up to 100x vs atlas methods.

Master's Researcher ([link](#))

Sep 2018 - Jun 2019

University of Sheffield

Sheffield, United Kingdom

Explored the training of classical and deep reinforcement learning agents on neuromorphic hardware.

- Developed Q-learning and SARSA classical and deep reinforcement learning agents in OpenAI Gym.
- Demonstrated that neuromorphic direct feedback alignment training performs similarly to backpropagation.

Research Engineer (Full-Time Internship) ([link](#))

Jun 2017 - Sep 2018

Rolls-Royce Holdings Plc.

Derby, United Kingdom

Modelled regional aviation market dynamics, investigating the feasibility of hybrid-electric regional aircraft.

- Adapted an internal agent-based transportation model to simulate regional air and ground travel dynamics.
- Tested aircraft designs under scenarios incorporating regulatory, noise, legal, and route-economics constraints.
- Delivered a strategic study to the CTO and external partners on long-term commercial opportunities.

Projects

- **Cleaning Robot API** ([link](#)): Dockerised FastAPI service with greedy A*-Manhattan planning and SQLite history.

Skills

- **Programming:** Python, C/C++ (familiar), Bash, MATLAB, SQL (familiar), LaTeX;
- **Deep Learning:** PyTorch, TensorFlow, TFLite, Keras, ONNX, Scikit-Learn, Statsmodels, OpenAI Gym;
- **Data Science:** NumPy, SciPy, Pandas, Matplotlib, Seaborn, MongoDB, HDF5;
- **Embedded:** TinyML, Digital Signal Processing, Optimisation, Quantisation, Emulation, Benchmarking;
- **Infrastructure:** Azure, Google Cloud, Docker, FastAPI, Slurm/HPC, Git, CI/CD, Nuitka, pytest;
- **AI-Assisted Engineering:** Codex, Claude Code, coding-agent workflows, specification-driven development;
- **Product Management:** Agile, Kanban, Cross-Functional Collaboration, Roadmapping, Design Thinking;
- **Business Development:** Technical Marketing, Horizon Scanning, Research Translation, Workshops, Pitching;

Select Certifications

- Digital Product Management Specialisation • *University of Virginia Darden School of Business* • ongoing
- AI for Medicine, Deep Learning, Machine Learning Specialisations • *Coursera* • 2018-2020

Publications and Select Talks

Publications

- Roibu, Andrei-Claudiu, et al. "Brain ages derived from different MRI modalities are associated with distinct biological phenotypes." *2023 10th IEEE Swiss Conference on Data Science (SDS)*. IEEE, 2023. ([link](#))

Talks, Workshops, and Pitches

- [Selling the Invisible: A Data-Driven Approach for Enabling In-Memory Computing for Edge AI](#) • SDS 2026
- [Unleashing In-Memory Computing with Synthara's ComputeRAM](#) • *Embedded World 2025*
- [Data Science Unmasks Brain Ageing: A Multimodal MRI Study](#) • *IEEE Swiss Conference on Data Science 2023*

Awards

- Royal Commission for the Exhibition of 1851 Industrial Fellowship (2021) ([link](#))
- Mappin Medal and Premium (Top Graduating Student at the University of Sheffield, 2019)

Additional Information

- **Languages:** Romanian (Native), English (C2), French (A2), German (A2)
- **Interests:** Reading (business, management, EQ), STEM outreach, Sports (trekking, jogging, kayaking, cycling)